

What’s next in AI?

Field notes from Microsoft Research for 2026

The story of AI in 2025 isn’t a tale of marginal gains. It is a narrative of scale and audacity.

What began as algorithms that nudged and assisted customers has evolved into systems that reason and adapt while collaborating with them. At Microsoft Research, the conversation around AI has moved beyond what’s possible to **what’s next**.

Across our global network of labs, researchers are rethinking the foundations of computing and intelligence. They are designing systems that govern themselves, embedding autonomy into the architecture of the digital world. They are building AI tools that work in low-resource languages and contexts, creating pathways for inclusion and access. They are advancing models that reason and understand human intent, and they are bringing intelligence into the physical realm, where robots learn and act with the fluidity of language.

BLOG

[Microsoft Source blog](#) 

These perspectives are drawn from Microsoft researchers, visionaries who are shaping the next frontier. They offer glimpses into what excites and challenges leaders at the edge of discovery and represent a series of expectations and ambitions for the year ahead. These voices are part of a larger conversation and complement a [Microsoft Source blog](#) in which the trends shaping AI in 2026 are framed, and it reflects a unified, One Microsoft strategy to set the agenda for the year ahead.

The ideas reflected below mark a change in how intelligence is imagined and applied. This is not about grafting AI onto old frameworks; it’s about reconstructing the core principles that drive progress itself. Microsoft is helping to shape that transformation and setting the direction for what comes next.

These are focal points from the edge of innovation, drawn from the work of Microsoft researchers across the globe. They trace the arc of progress and hint at the possibilities that will define the next chapter of AI.

AI as a lab assistant helping to accelerate scientific discovery



“AI will join in the process of discovery, creating a world where every research scientist has AI lab assistants that suggest and run parts of experiments.”

– [Peter Lee](#), *President, Microsoft Research*

AI is already speeding up the [pace of scientific discovery](#). Researchers are using AI across areas like climate modeling, molecular dynamics, materials design, and more. But AI will do more than just model physics, chemistry, and biology. And it won’t be limited to summarizing papers, answering questions, and writing reports.

> [Read more from Peter](#)

Autonomous agents will transform digital economies



“We stand at the threshold of a new economic era—one where autonomous agents collaborate, negotiate, and transact on behalf of people and organizations.”

– [Saleema Amershi](#), *Partner Research Manager, Microsoft Research AI Frontiers*

As AI agents evolve from isolated tools to active participants in our digital ecosystems, we stand at the threshold of a new economic era—one where autonomous agents collaborate, negotiate, and transact on behalf of people and organizations. These agentic ecosystems promise to reorganize digital marketplaces, reduce friction, and broaden access to opportunity. Realizing this vision requires rethinking the architecture of systems, platforms, and protocols that underpin digital markets with an agent-native lens.

> [Read more from Saleema](#)

AI meets biology to decode life’s language



“Biology stores this incredible scale, richness and complexity of data within each and every one of us—and today we’re leveraging AI to decode that language to design new biomolecules and discover mechanisms of disease.”

– [Ava Amini](#), *Principal Researcher, Microsoft Health Futures*

For decades, computational biology operated within narrow lanes, predicting protein structures or analyzing gene expression, while medicine largely treated patients as averages. What’s new is the rise of generative AI models that treat biology as a language, enabling systems to design new proteins and predict cellular behaviors that can lead to personalized therapies.

> [Read more from Ava](#)

Future AI infrastructure enabling the next 1,000x



“Light-based chips and robotics-enabled data-center designs promise an era where AI infrastructure is faster, more sustainable, more reliable, and fundamentally different.”

– [Hitesh Ballani](#), Partner Research Manager, Microsoft Research Cambridge

In 2026, two forces will redefine AI infrastructure. First, [AI-driven system intelligence](#) will unlock a step change in efficiency and velocity through automated tooling for developing, deploying, and optimizing models, all co-designed with the underlying hardware. Early signals are already here, with tooling accelerating adoption of [models optimized for the edge](#).

> [Read more from Hitesh](#)

Scaling AI at the speed of light



“We could move toward smaller, more efficient compute modules paired with shared memory pools, all connected through a fast, unified, low-power optical fabric.”

– [Paolo Costa](#), Partner Research Manager, Microsoft Research Cambridge

AI scaling is entering a new chapter. The challenge involves moving data quickly across GPUs and between GPUs and memory without burning too much energy. But new solutions are emerging. Throughout 2025, we’ve seen advances in low-power, high-bandwidth optical interconnects, from [our own microLED work](#) to developments across the [GPU](#) and [networking](#) ecosystem. As these technologies mature, I expect 2026 to be a pivotal year in moving them from R&D to early deployments, with wider adoption by decade’s end.

> [Read more from Paolo](#)

AI that amplifies human agency through inclusive innovation



“Imagine learning assistants that understand current learning levels and styles, the local context, curricula, and languages, and use this information to navigate the best learning path.”

– [Tanuja Ganu](#), Director of Research Engineering, Microsoft Research India

AI’s next frontier is not just smarter algorithms and models but the systems that amplify human agency in high-stakes domains like education, agriculture, and healthcare. The challenge: design AI-native workflows that serve a teacher or student in rural India, a farmer in Kenya, or a frontline health worker in Brazil. The answer lies in building AI that closes opportunity gaps and creates pathways for empowerment rather than reinforcing existing divides.

> [Read more from Tanuja](#)

From reasoning to simulating and mentalizing



“The AI community is now shifting its research focus from merely encoding world knowledge through large language models to developing reasoning abilities by enabling AI models to interact with their environments.”

– [Jianfeng Gao](#), Distinguished Scientist, Microsoft Research Redmond

Every groundbreaking scientific revolution that transforms our lives begins with asking profound questions. Inquiries about the fundamental components of the universe, the measurement of information, and the origins of life have led to the discovery of three essential elements that define our physical and digital worlds: the atom, the bit, and the gene, respectively.

> [Read more from Jianfeng](#)

Spatial intelligence moves from perception to action



“Digital agents will not only perceive the world, but understand, predict, and act within it.”

– [Baining Guo](#), Distinguished Scientist, Microsoft Research Asia

Spatial intelligence is entering a transformative new era—one where digital agents will not only perceive the world, but understand, predict, and act within it. As physical and virtual environments continue to merge, a set of powerful technological trends is reshaping what intelligent systems can do.

> [Read more from Baining](#)

Audience-shaped stories in real time



“This convergence of AI and entertainment will redefine how we connect, play, and create – not as passive consumers, but as active participants in living, breathing worlds.”

– [Katja Hofmann](#), Partner Research Manager and lead of the Microsoft Research Game Intelligence team

Looking ahead, AI is poised to transform entertainment into deeply interactive and personalized experiences. We’ll move beyond static content toward dynamic worlds that adapt in real time to player choices, emotional context, and even social interactions. Generative models will empower creators to co-design with AI, blending narrative, art, and technology in ways that feel truly alive. Imagine stories that evolve with audience feedback, characters that learn and grow over time, and immersive environments that respond to human creativity and curiosity.

> [Read more from Katja](#)

AI as a partner to stabilize workflows and accelerate innovation



“Agentic systems will hold context across months, track evolving goals, surface forgotten assumptions, and help teams stay oriented in the messy middle of innovation.”

– [Weishung Liu](#), Principal Program Manager, Microsoft Research Redmond

In 2026, AI stops acting like a tool you query and starts behaving like a collaborator you build momentum with. The breakthrough that enables this shift will be memory. Agentic systems will hold context across months, track evolving goals, surface forgotten assumptions, and help teams stay oriented in the messy middle of innovation. This reduces the cognitive tax of complex work and makes iteration feel continuous rather than fragmented. With AI as a stabilizing partner, teams will be able to move faster, and organizations will be able to preserve the intent behind their best discoveries, not just the artifacts.

Adaptive and collaborative robotics



“What’s new is the emergence of AI ‘vision-language-action’ models for physical systems that will soon perceive, reason, and act in dynamic environments alongside humans.”

– [Ashley Llorens](#), Corporate Vice President and Managing Director, Microsoft Research Accelerator

Physical AI, where agentic AI meets physical systems, is poised to redefine robotics in the same way generative AI has transformed language and vision. For decades, robots have excelled in structured environments like assembly lines and warehouses, where tasks are predictable and tightly scripted. What’s new is the emergence of AI “vision-language-action” models for physical systems that will soon perceive, reason, and act in dynamic environments alongside humans. These models translate natural language prompts into physical behaviors, enabling robots to adapt their actions by generalizing experiences across familiar scenarios with subtle variations rather than failing the first time they are presented with unexpected situations.

> [Read more from Ashley](#)

The many ways AI will reshape communication



“Communication will increasingly unfold as an iterative process, more closely aligned with how human thinking naturally progresses.”

– [Yan Lu](#), Assistant Managing Director, Microsoft Research Asia

Today, AI systems can generate text, images, and videos at an unprecedented scale, making information more accessible than ever. Yet the human capacity to interpret and internalize this growing stream of information is limited, creating a widening gap between content generation and comprehension.

> [Read more from Yan](#)

Improved context management helps AI act, not just respond



“Agents will generate and consume far more information than a single prompt can hold, so ‘context engineering’ will become essential.”

– [Dasha Metropolitansky](#), *Research Data Scientist, Microsoft Research Redmond*

The next wave of AI will go far beyond question-answering and content generation. Agents will perform increasingly complex, long-running tasks that rely on external tools and real-time data. This shift introduces a critical challenge: preserving coherence and direction over extended sequences of actions. Agents will generate and consume far more information than a single prompt can hold, so “context engineering” will become essential, dynamically curating and structuring instructions, tools, and memories to keep the system on track. Improved state management will help today’s promising agents become tomorrow’s dependable collaborators.

AI empowering billions by turning vision into reality



“AI can help humans scale, and humans can ensure that the AI stays true to the intended purpose.”

– [Venkat Padmanabhan](#), *Managing Director, Microsoft Research India*

AI has seen breathtaking growth in capabilities and adoption. It is important to also take a global perspective, and we’re taking a three-pronged approach.

> [Read more from Venkat](#)

The language of the patient as a pathway to precision health



“Progress depends on learning the language of the patient ... Each modality shouts insights that sound like noise unless we understand the multimodal language behind them.”

– [Hoifung Poon](#), *GM, Real-World Evidence, Microsoft Research Redmond*

Today, each of us enjoys healthcare far superior to what any king or queen could have imagined centuries ago. That raises a profound question: can we enable every person on the planet to access care better than the most resourced individual today? With the rise of generative AI, we are beginning to knock on that door.

> [Read more from Hoifung](#)

Psychological well-being as a core element of AI design and governance



“Psychological stewardship must be multidimensional, safeguarding individual resilience, fostering trust and empathy in interpersonal dynamics, and reinforcing societal cohesion.”

– [Jina Suh](#), Principal Researcher, Microsoft Research Redmond

As AI systems permeate everyday lives, their influence on psychological well-being is no longer theoretical. AI is mediating decisions, relationships, and narratives across personal, work, education, and civicspheres, and shaping how people think, feel, behave, relate, and understand themselves.

> Read more from Jina

AI as a trusted companion in society



“We anticipate a major shift in how AI interacts with people, not as a tool that executes tasks, but as a trusted companion that collaborates, reasons, and grows alongside us.”

– [Xing Xie](#), Assistant Managing Director, Microsoft Research Asia

In 2026, we anticipate a major shift in how AI interacts with people, not as a tool that executes tasks, but as a trusted companion that collaborates, reasons, and grows alongside us. Instead of focusing only on accuracy or efficiency, AI systems will behave in ways consistent with human expectations for cooperation, respect, and social intuition.

> Read more from Xing

Multimodal foundation models and agentic systems for patient care



“Instead of only drafting summaries, future agentic systems may support triage, diagnostics, treatment planning, and coordinated follow-up.”

– [Xinxing Xu](#), Principal Research Manager, Microsoft Research Asia

The next phase of AI for healthcare will be defined by multimodal and agentic capabilities. Multimodal foundation models are beginning to integrate text (such as clinical notes), medical images, physiological signals, and genomics into a unified representation space for enhanced patient care. This enables systems to move beyond narrow clinical diagnoses toward broader, context-rich clinical reasoning.

> Read more from Xinxing

System intelligence as the next leap



"... we must define and measure what intelligence in systems truly means, capturing how AI reasons about architectures, trade-offs, and correctness."

– [Lidong Zhou](#), Corporate Vice President and Lab Director, Microsoft Research Asia

The future of computing isn't just faster—it's smarter. [System Intelligence](#) moves beyond AI's ability to generate code toward designing, optimizing, and governing entire systems. Instead of static architectures, imagine infrastructures that adapt and evolve toward high-level goals.

> [Read more from Lidong](#)



Story contributors: Courtney Diamond, Kate Forster, David Celis Garcia, Lindsay Kalter, Amanda Melfi, Brenda Potts, Lindsay Shanahan, Amber Tingle

Explore more

- [Microsoft Research Podcast](#)
- [Microsoft Research Blog](#)
- [Microsoft Research Focus newsletter](#)

- [Microsoft Research AI Frontiers](#)
- [Microsoft Research Asia](#)
- [Microsoft Research Cambridge](#)
- [Microsoft Health Futures](#)
- [Microsoft Research Redmond](#)

Follow us:      

Share this page:    

What's new	Microsoft Store	Education	Business	Developer & IT	Company
Surface Pro	Account profile	Microsoft in education	Microsoft AI	Azure	Careers
Surface Laptop	Download Center	Devices for education	Microsoft Security	Microsoft Developer	About Microsoft
Surface Laptop Studio 2	Microsoft Store support	Microsoft Teams for Education	Dynamics 365	Microsoft Learn	Company news
Copilot for organizations	Returns	Microsoft 365 Education	Microsoft 365	Support for AI marketplace apps	Privacy at Microsoft
Copilot for personal use	Order tracking	How to buy for your school	Microsoft Power Platform	Microsoft Tech Community	Investors
AI in Windows	Certified Refurbished	Educator training and development	Microsoft Teams	Microsoft Marketplace	Diversity and inclusion
Explore Microsoft products	Microsoft Store Promise	Deals for students and parents	Microsoft 365 Copilot	Marketplace Rewards	Accessibility
Windows 11 apps	Flexible Payments	AI for education	Small Business	Visual Studio	Sustainability



[Your Privacy Choices](#)

[Consumer Health Privacy](#)

[Sitemap](#)

[Contact Microsoft](#)

[Privacy](#)

[Terms of use](#)

[Trademarks](#)

[Safety & eco](#)

[Recycling](#)

[About our ads](#)

[© Microsoft 2026](#)